

Passive photodiode visibility — run report

RUN

214942_pd-visibility_pd-visibility

Status: failed | Started: 2026-09-21T20:49:42.950651+00:00

PURPOSE

Measure fringe contrast on the selected PD input under external drive. No RP output initialization or PAX connection.

REQUIRED OPTICAL SETUP

DC-coupled PD on the selected RP input; external drive must span complete fringes. Set its frequency and, when known, the blocked-light voltage offset. RP must already have the Pyrpl FPGA loaded.

RECORDED BENCH CONTEXT (RECIPE)

In front of NPBS 1, C port

Red Pitaya (0-1 V command) -> Thorlabs MDT690 -> piezo

DATA

0 recorded samples

Source: data.csv; full acquisition settings: recipe.json; status: run.json

RUN ISSUE

ValueError: Capture must contain at least two external-drive periods

Run log

RECORDED OUTPUT

WARNING:pyrpl.modules:Register decimation of module scope has value 16384, which does not correspond to selected option 8192. Setting to '8192'.

Passive visibility on in2; external drive 0.5 Hz; RP outputs untouched.

DC coupling and a full fringe excursion are required. Extrema use 1/64-period bin averages.

Dark voltage not measured: reporting apparent contrast assuming zero offset.

WARNING:pyrpl.modules:Trying to load attribute decimation of module scope that are invalid setup_attributes.

Run failed: ValueError: Capture must contain at least two external-drive periods

Active parameters

RP commands: volts, bounded to 0–1. V_lambda: terminal volts for 2π . Gains: terminal V / RP V.

Parameter	Recorded value
label	"pd-visibility"
duration_s	12.0
report	"pdf"
pd_input	"in2"
visibility_frequency_hz	0.5
visibility_dark_voltage_v	null